

Bachelor Level/ Second Year/ Third Semester/ Science
Computer Science and Information Technology (CSc. 203)
(Operating System)

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Section A

Attempt any two questions:

(2x10=20)

1. What is System Calls? Explain the system call flow with the help of a block diagram.

OR

What do you mean by file systems? What are the major difference between file system interfaces & file system implementation? Explain.

2. Write short notes on:

- a) Disk Scheduling Algorithms.
- b) Error Handling & Formatting.
- c) File Operations.

3. Consider the following page reference string : 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. How many page faults would occur for the LRU replacement, FIFO replacement & optimal replacement algorithms? Assuming three, five or seven frames? Remember all frames are initially empty, so your first unique pages will all cost one fault each.

Section B

Attempt any eight questions:

(8x5=40)

4. Differentiate between personal computer operating systems & mainframe operating systems.
5. When do page fault occur? Describe the actions taken by an OS when a page fault occurs.
6. List four necessary conditions for deadlock. Explain each of them briefly what would be necessary (in the operating system) to prevent the deadlock.
7. Draw & describe the 3-state process model.
8. Does window have any concept of process hierarchy? How does parent control the child?
9. What is the problem with thread implementation in user space when any of the threads get blocked while performing I/O operation?
10. Explain why two level & scheduling is commonly used.
11. What are the main motivations & issues in primary memory management?
12. Explain the disk management with example.